

HAYAT MEDICAL COLLEGE

Progressive assessment II for Year I Medical Students

Date: December 19, 2014

Time allowed: 3:00 hours.

Name: _____

HISTOLOGY

MCQ: Each of the numbered items or incomplete statements is followed by answers or by completions of the statement. Select the one lettered answer or completion that is best in each case. (Total : 50 points)

_____1. Concerning the **teeth and associated tissues**, which statement is **true**?

- a) Dentin is more calcified than enamel.
- b) Pulp is composed of dense connective tissue.
- c) Enamel is replaced in the root by cementum.
- d) Odontoblasts make enamel.
- e) None of the above.

_____2. Which of the following statements is **false** in the **oral cavity**?

- a. Taste buds occur most commonly on the sides of fungiform and circumvallate papillae.
- b. The whitish tips of filiform papillae consist of keratinized epithelial tissue.
- c. Sublingual gland lacks serous acini but contains serous demilunes.
- d. Periodontal ligament is composed of mucoid connective tissue.
- e. Serous demilunes produce lysozyme.

_____3. Regarding the **stomach**, which statement is **true**?

- a) Is able to withstand the acidic environment because it is lined by stratified squamous epithelium.
- b) Has its gastric glands best developed in the pylorus than the other regions.
- c) Forms temporary folds called plicae circulares when it is empty.
- d) Has its muscularis externa made by skeletal muscle fibres.
- e) Mucosal glands of the pylorus secrete mainly mucus.

_____4. Concerning the **stomach**, which statement is **true**?

- a. It is specifically responsible for the absorption of vitamin B₁₂ and the reabsorption of conjugated bile salts.
- b. Goblet cells are more abundant to produce mucus as well as appreciable amounts of lysozyme.
- d. Hydrochloric acid and intrinsic factors are secreted by enteroendocrine cells of cardia and pylorus.
- d. Parietal cells also called oxyntic cells contain mucinogen granules and pepsinogen.
- e. The folds of the empty stomach are called rugae.

_____5. Which one of the following marks best the **ileum** from the **duodenum** and **jejunum**?

- a) Long and broad leaf like villi.
- b) Absence of taenia coli.
- c) Peyer's patches.
- d) Plicae circulares.
- e) Well developed intestinal glands.

_____6. With regard to the **small intestine**, which statement is **true**?

- a. The muscularis externa in the upper third of duodenum and jejunum is composed of striated muscle.
- b. Crypts of Lieberkühn are lymphatic capillaries [lacteals] within a villus of the small intestine.
- c. Ileum characteristically contains the most goblet cells in the mucosa covering its villi.
- d. Plicae circulares are projections with a core of mucosa.
- e. Villi are finger like projections of submucosa.

- 7. **Diarrhea** may result if which of the following organs fails to carry out its primary role in **absorbing water from the feces**?
- a) anal canal b) colon c) cecum d) small intestine e) stomach
- 8. With regard to the **large intestine**, which statement is **false**?
- a. Lacks villi and contains a disproportionately large number of goblet cells.
b. The large intestine is shorter than the small intestine but has a wide lumen
c. The muscularis externa has three thick bands of muscle called the tenia coli.
d. The lamina propria has many macrophages, plasma cells, eosinophils, and lymphoid nodules.
e. Paneth cells of the submucosal glands house bacteria that produce vitamin B₁₂ and vitamin K.
- 9. Which of the following statements is **correct** regarding the **intestine**?
- a) The striated border of the small intestine is made by microvilli. b) Small intestine has tenia coli.
c) The absorptive cells of the small intestine undergo mitosis at the tips of the villi and are sloughed into the crypts. d) Large intestine possesses branched alveolar glands.
e) Brünner's glands in mucosa of the duodenum produce mucin containing alkaline fluid and the polypeptide hormone urogastrone and empty their secretions into the lacteals.
- 10. Concerning the **Vermiform appendix**, select the **correct** statement.
- a) has few lymphoid nodules in the lamina propria and submucosa.
b) is a blind-ended evagination of the jejunum. c) has relatively shallow intestinal glands.
d) longitudinal smooth muscle gathers into its teniae coli. e) contains short villi.
- 11. Which is **not the function of the liver**?
- a. Storage of glycogen, fat soluble vitamins, iron and cholesterol.
b. Detoxification/removal of drugs and alcohol. c. Storage of calcium and bile.
d. Production of plasma proteins and lipoproteins. e. Fetal RBC production.
- 12. With regard to the **pancreas**, which statement is **false**?
- a. Islets of Langerhans are incompletely delimited from the acini by Schwann cells.
b. Myoepithelial cells of the striated ducts deliver bicarbonate ions into the primary secretion.
c. The exocrine portion is a compound tubuloacinar gland, similar in structure to the parotid gland.
d. It is a lobulated retroperitoneal gland whose islet cells are compact mass of epithelial cells not cords or plates.
e. Although it is a mixed exocrine and endocrine gland, the majority of the secretory cells are exocrine cells.
- 13. Which structure has a **mucosa lined by nonkeratinized stratified squamous epithelium**?
- a. Esophagus b. Stomach c. Small intestine d. Large intestine e. Rectum
- 14. Concerning the **GI tract**, select the statement which is **not correct**.
- a. The muscularis externa consists of Auerbach's plexus.
b. Hepatic sinusoids are surrounded by the space of Disse.
c. Masticatory mucosa is located on the hard palate and gingiva.
d. The mucosa of the gallbladder is made of simple columnar epithelium.
e. The adventitia consists of only a single layer of epithelial cells and underlying connective tissue.
- 15. With regard to the **GI tract**, select the statement which is **correct**.
- a. Permanent circular folds (plicae circulares) are characteristic structural features of the esophagus.
b. Fat-storing stellate cells (Ito cells) in the space of Disse in the liver are noted for storing vitamin A.
c. Myenteric (Auerbach's) plexus is located just beneath the epithelium of the mucosa, in lamina propria.
d. A hepatic cord is the tough connective tissue surrounding a portal triad.
e. Like the small intestine, the gall bladder has villi with lacteals adjacent to the Paneth and goblet cells in the mucosal epithelium.

————16. **Transitional epithelium** (urothelium) lines all of the following **EXCEPT** the:
a. Bowman's capsule. b. renal pelvis. c. ureter. d. bladder. e. proximal urethra in males.

————17. Which of the following is characteristic of the **proximal** but **not** of the **distal convoluted tubules**?
a) convoluted portion lies in the renal cortex b) epithelial lining is similar to that of the loop of Henle
c) exhibits a brush border d) lining cells are basophilic
e) secretes glucose and amino acids into the filtrate

————18. Which of the following is **not** a **function of the kidney**?
a. Erythropoietin production b. Vitamin D modification c. Acid base balance
d. Aldosterone production e. Renin production

————19. Which of the following is **not** the function of **mesangial cells**?
a. synthesis of extracellular matrix. b. endocytosis and contractility.
c. maintenance of acid-base balance in the blood. d. structural support to the glomerulus.
e. increase blood flow and the effective surface area available for ultrafiltration.

————20. In the **urinary system**, which statement is **false**?
a. Aldosterone acts on the distal convoluted duct.
b. Peritubular capillaries arise from the efferent arterioles.
c. The renal pyramid and its associated cortex is referred to as the renal lobe.
d. Polyploidy and binucleated cells are common among cells of the transitional epithelium.
e. Renin is secreted by podocytes of the visceral layer of Bowman's capsule referred to as area cribrosa.

————21. In the **urinary system**, which statement is **true**?
a. Interstitial cells in the renal medulla secrete erythropoietin through filtration slits.
b. Macula densa are closely-packed nuclei in the epithelium of the proximal convoluted tubule.
c. Endothelial cells of glomerular capillaries are of non-fenestrated variety.
d. The distal convoluted tubules possess neither brush borders nor apical canaliculi.
e. The cortical collecting tubules are located primarily at the apex of the renal pyramids and are lined by pseudostratified ciliated columnar cells.

————22. Epithelial cells lining the **thin limb of the loop of Henle**:
a. are modified smooth muscle cells responsible for the secretion of rennin.
b. have an apical brush border of microvilli for active absorption from the lumen.
c. contain many ribosomes and rough endoplasmic reticulum for extensive protein synthesis.
d. contain numerous basally located secretory vesicles. e. are simple squamous.

————23. **Testis** has the following coverings **except**:
a. tunica adventitia b. tunica vasculosa c. tunica albuginea d. tunica vaginalis e. None

————24. In the **testis**, which statement is **true**?
a. The seminiferous tubules are lined with a complex stratified epithelium.
b. The secondary spermatocytes are the largest cells of the spermatogenetic lineage.
c. The tunica vaginalis is thickened on the posterior surface to form the mediastinum testis.
d. Interstitial cells (of Leydig) produce testicular fluid which constitute the bulk of ejaculate volume.

e. During spermiogenesis, the spermatids quickly undergo the second meiotic division to give spermatozoa.

—————25. Which of the following is **NOT** considered part of the **male genital duct system**?

- a. Rete testis b. Tubuli recti c. Seminal vesicles d. Ductus deferens e. Ductus epididymidis

—————26. Which of the following statements is **false** in the **male reproductive system**?

- a. The secretion of seminal vesicle is particularly rich in fructose.
b. Secondary spermatocytes are smaller than the primary spermatocytes.
c. The epithelium lining the ductus epididymis is pseudostratified columnar.
d. Tunica albuginea is the thickest in epididymis. e. The subcutaneous tissue in penis is devoid of fat.

—————27. The function of **Sertoli cells** includes all of the following **EXCEPT**:

- a. phagocytosis of residual bodies from maturing spermatids.
b. establishment through tight junctions of a blood/testis barrier.

c. They coagulate semen and form vaginal plug.

- d. metabolic and mechanical support of germ cells. e. secretion of tubular fluid.

—————28. Which of the following statements is **true** with regard to the **male reproductive system**?

- a. A network of tubules within the testis which interconnects the seminiferous tubules with the efferent ductules is the epididymis.
b. Epithelial lining of the duct of epididymis is columnar epithelium with brush borders and goblet cells.
c. The ejaculatory duct connects the epididymis with the seminal vesicle where sperm cells mature after spermiogenesis.
d. Semen contains abundant fluid derived from the lymphatics lined by mesothelium of the corpus spongiosum and corpora cavernosa.
e. The seminal vesicle consists of a hollow tube with an extensively folded mucosal lining, in a fibromuscular stroma.

—————29. Which of the following statements is **false** in the **male reproductive system**?

- a. Ductus epididymidis secretes glycerophosphocholine.
b. The penile urethra passes through the corpus spongiosum.
c. The secretion of seminal vesicle is particularly rich in fructose.
d. Spermatozoa are motile during storage in the epididymis, but become nonmotile in the ejaculate.
e. The prostate gland, the largest accessory structure, is composed of many compound tubuloalveolar glands.

—————30. Which of the following statements is **true** about the **female reproductive system**?

- a. Mucosa of the uterus is commonly called perimetrium.
b. Fornix of vagina is lined by stratified squamous epithelium.
c. The life span of corpus luteum of pregnancy is about 10 lunar months.
d. Primary oocyte completes its first meiotic division at the time of fertilization.
e. The acidic pH in the vagina is due to the secretion of HCl by the tubuloalveolar glands of the vaginal walls.

—————31. Which of the following statements is **true** in the **female reproductive system**?

- a. The cells of theca interna secrete progesterone.
b. Germinal (surface) epithelium covering the ovary gives rise to ova.

- c. Glassy membrane is the last structure to disappear in an atretic follicle.
- d. Corpora albicans or "*white bodies*" are ovarian follicles which degenerate after failing to mature.
- e. After ovulation, granulosa cells and cells of theca interna grow and differentiate to form the primordial follicles.

_____32. Which of the following is true of the **ampulla** of the **uterine tube** but not of its **isthmus**?

- a) Contains mucus-secreting peg cells.
- b) Contains large mucosal folds that are extensively branched.
- c) Contains smooth muscle in its walls.
- d) Is covered by serosa derived from the peritoneum.
- e) Contains ciliated epithelial cells.

_____33. Which of the following statements is **true** in the **female reproductive system**?

- a. During ovarian follicle maturation, surrounding stromal cells differentiate to form the zona pellucida.
- b. The primary follicles are dependent on FSH for their development.
- c. Primary oocyte completes its first meiotic division at the time of ovulation.
- d. Primordial follicle is identified in section by the presence of cumulus oophorus.
- e. Cilia are found on the free surface of corona radiata cells.

_____34. **Liquor folliculi**,

- a) stimulates the myoepithelial cells of the mature follicle to rupture.
- b) acts as a porous filter through which dissolved hormones and enzymes are prevented.
- c) causes the endometrial glands to secrete and prepare the endometrium for implantation.
- d) is composed of cumulus oophorus and polar bodies.
- e) None of the above.

_____35. In the **female reproductive system**, which of the following statements is **false**?

- a. Mammary gland is a modified apocrine sweat gland.
- b. Underneath the germinal epithelium of the ovary is connective tissue called the tunica albuginea.
- c. Proliferative phase of uterus coincides with the development of ovarian follicles.
- d. In the ovaries, the medulla harbors most of the oocyte-containing ovarian follicles.
- e. during pregnancy the endometrium grows extensively by hypertrophy and hyperplasia.

_____36. In the **female reproductive system**, which of the following statements is **correct**?

- a. Zona pellucida is the connective tissue layer around the primary follicle.
- b. The difference in sex chromosome complement of oocytes forms the basis of primary sex determination.
- c. the perimetrium is a trumpet-shaped opening towards the peritoneal cavity.
- d. the basal layer of endometrium is sloughed off during each menstruation.
- e. Glassy membrane is the last structure to disappear in an atretic follicle.

EMBRYOLOGY

MCQ: Each of the numbered items or incomplete statements is followed by answers or by completions of the statement. Select the one lettered answer or completion that is best in each case. (Total : 50 points)

- _____ 1. The **cardiovascular system** is not derived from:
- a). Splanchnic mesoderm
 - b). Endoderm
 - c). Paraxial mesoderm
 - d). lateral mesoderm
 - e). Neural crest cells
- _____ 2. The **heart primordium** between the fourth and seventh weeks **does not** consist of:
- a. the bulbus cordis
 - b. ventricle
 - c. atrium
 - d. sinus venosus
 - e. auricles.
- _____ 3. Concerning the **development of the heart**, which statement is true?
- a) The bulbus cordis forms the trabeculated part of the right ventricle.
 - b) The prevalence of atrial septal defects is more common in females than in males.
 - c) The left horn of sinus venosus becomes incorporated into the wall of the right atrium.
 - d) Sinoatrial nodes are derived from cells of the atrioventricular canal during the 8th week.
 - e) The truncus arteriosus forms the roots and proximal portion of the aorta and pulmonary artery.
- _____ 4. With regard to the **development of cardiovascular system**, which statement is **false**?
- d). Cavitation of the ventricular walls forms the trabeculae carneae.
 - a). The visceral pericardium is derived from mesothelial cells of the bulbus cordis.
 - b). The right anterior cardinal vein and the right common cardinal vein become the SVC.
 - e). The transverse pericardial sinus forms from the breaking away of the dorsal mesentery.
 - c). The foramen primum serves as a shunt, enabling oxygenated blood to pass from the right to the left atrium.
- _____ 5. With regard to the **development of cardiovascular system**, which statement is **false**?
- a. Coarctation of the aorta occurs twice as often in males as in females.
 - b. Umbilical veins carry well-oxygenated blood from the chorion.
 - c. Ventricular septal defects (VSDs) occur more frequently in males than in females.
 - d. Persistent ductus arteriosus (PDA) is two to three times more frequent in females than in males.
 - e. The distal part of the right sixth pharyngeal arch artery forms a prenatal shunt, the ductus arteriosus.
- _____ 6. With regard to the **development of cardiovascular system**, which statement is **false**?
- a. Most of the intra-abdominal parts of the umbilical arteries become the ligamentum venosum.
 - b. The right ventricular wall is thicker than the left ventricular wall in fetuses and newborn infants.
 - c. The intra-abdominal part of the umbilical vein eventually becomes the round ligament of the liver.
 - d. The short, thick ligamentum arteriosum extends from the left pulmonary artery to the arch of the aorta.
 - e. None of the above.
- _____ 7. Select the **incorrect** statement pertaining to the **development of the aortic arches**.
- a. The right fourth aortic arch becomes the right pulmonary artery.
 - b. Dorsal parts of the fifth pair of aortic arches form the ductus arteriosus.
 - c. The right fourth pharyngeal arch artery becomes the proximal part of the right subclavian artery.
 - d. First pair of pharyngeal arch arteries may contribute to the formation of the external carotid arteries.
 - e. Ventral parts of the second pair of pharyngeal arch arteries persist and form the stems of the maxillary

arteries.

- _____ 8. With regard to the **development of the arterial system**, which statement is **false**?
- a) the external carotid artery is a sprout of the 4th aortic arch.
 - b) the mesenteric arteries are derived from the right proximal umbilical arteries.
 - c) the aortic sac forms right and left horns which subsequently gives rise to the brachiocephalic artery.
 - d) the 1st aortic arch gives rise to stapedial arteries.
 - e) All of the above.

_____ 9) Aortic arch IV forms the _____ on the right side of the embryo and the _____ on the left.

- a) Right subclavian artery; Arch of aorta
- b) Ductus arteriosus; Pulmonary Artery
- c) Arch of aorta; Pulmonary Artery
- d) Ductus arteriosus; Right subclavian artery
- e) Right subclavian artery; Left subclavian artery

- _____ 10. Which of the following is not the classic group of cardiac defects of the **Tetralogy of Fallot**?
- a. Pulmonary stenosis (obstruction of right ventricular outflow)
 - b. Ventricular septal defect
 - c. Dextroposition of aorta (overriding or straddling aorta)
 - d. Left ventricular hypertrophy
 - e. Right ventricular hypertrophy

- _____ 11) During week six (lymphatic vessel development), the dilated lymph sac that forms at the lower end of the thoracic duct is called the _____.
- a) Lymphocitic Sac
 - b) Iliac Lymph Sac
 - c) Retroperitoneal Lymph Sac
 - d) Jugular Lymph Sac
 - e) Cisterna Chyli

- _____ 12. Which of the following primitive veins gives the **hepatic segment of inferior vena cava**?
- a) Right supracardinal vein
 - b) Right subcardinal vein
 - c) Right vitelline vein
 - d) Right Umbilical vein
 - e) Right posterior cardinal vein

- _____ 13. With regard to the **development of the venous system**, which statement is **true**?
- a) the left subcardinal vein forms the azygos vein.
 - b) After birth umbilical vein becomes ligamentum venosum.
 - c) the anastomosis between the anterior cardinal veins forms the left renal vein.
 - d) the anastomosis between the subcardinal veins forms the left brachiocephalic vein.
 - e) the anastomosis between the sacrocardinal veins forms the left common iliac vein.

- _____ 14. The **urinary system** develops mainly from:
- a) Ectoderm
 - b) Paraxial mesoderm
 - c) Intermediate mesoderm
 - d) Lateral plate mesoderm
 - e) Endoderm

- _____ 15. Concerning the **development of the urinary system**, which statement is **false**?
- a) Urinary system begins to develop before the genital system.
 - b) The metanephric diverticulum is the primordium of the ureter.
 - c) Nephrons are derived from the metanephric mass of intermediate mesoderm.
 - d) Differentiation of the nephrons depends upon induction by the collecting tubules.
 - e) The kidneys attain their definitive adult position toward the end of the fetal period.

- _____ 16. During the **development of the Urinary System**, which statement is **true**?
- a) The urorectal septum divides the urachus from the allantois.
 - b) Bilateral renal agenesis is associated with oligohydramnios.
 - c) The podocytes of Bowman's capsule are derived from endoderm and ectoderm.
 - d) Urachal cyst is a remnant of the cloaca.
 - e) Urine production begins shortly after birth.

- _____ 17. With regard to the ***development of the urinary system***, which statement is ***false***?
- a. The metanephros become functional at week 10 of development.
 - b. The podocytes of Bowman's capsule are derived from mesoderm.
 - c. The transitional epithelium lining the urethra is derived from mesoderm.
 - d. Connecting tubule the adult derivative of the ureteric bud.
 - e. Urachal cyst is a remnant of allantois.

- _____ 18. Concerning the development of the ***urinary bladder*** and ***urethra***, which statement is ***false***?
- a) The female urethra is derived from endoderm of the urogenital sinus.
 - b) In adults, the urachus is represented by the median umbilical ligament.
 - c) The urinary bladder becomes a pelvic organ during the 9th week of development.
 - d) The trigone of the urinary bladder is derived from the caudal ends of the mesonephric duct.
 - e) The epithelium of the urinary bladder is derived from the endoderm of the vesical part of the urogenital sinus.

- _____ 19. Concerning the ***development of the urinary system***, which statement is ***false***?
- a) Urinary system begins to develop before the genital system
 - b) Nephrons are derived from the metanephric mass of intermediate mesoderm.
 - c) Differentiation of the nephrons depends upon induction by the collecting tubules.
 - d) The kidneys attain their definitive adult position toward the end of the fetal period.
 - e) The proximal convoluted tubules of the definitive adult kidney are derived from metanephric vesicle.

- _____ 20. With regard to the ***congenital anomalies of the urinary system***, which statement is ***false***?
- a) Accessory renal arteries are end arteries.
 - b) Bilateral renal agenesis is associated with oligohydramnios.
 - c) Unilateral renal agenesis is more common in females than males.
 - d) Incontinence is a common complaint resulting from an ectopic ureteric orifice in both sexes.
 - e) The "ascent" of the horse shoe kidney is prevented by the root of the inferior mesenteric artery.

- _____ 21. With regard to the ***development of gonads***, which statement is ***false***?
- a) The surface epithelium (germinal epithelium) of the female gonad develop into oogonia.
 - b) Sex of an embryo is determined genetically at the time of fertilization.
 - c) Germ cells do not appear in the genital ridges until the 6th week of development.
 - d) By the 8th week of gestation, the interstitial cells of Leydig begin to produce testosterone.
 - e) In embryo with an XY sex chromosome complex, medullary cords develop into the testis.

- _____ 22. With regard to the ***vestigial structures derived from embryonic genital system***, which statement is ***false***?
- a) The prostatic utricle is homologous to the vagina.
 - b) The cranial end of mesonephric duct in males may persist as paradidymis.
 - c) The epoophoron in the females correspond to the efferent ductules and duct of epididymis.
 - d) The cranial end of the paramesonephric duct in males may persist as a vesicular appendix of testis.
 - e) Part of the cranial end of the paramesonephric duct in females that does not contribute to the infundibulum of the uterine tube may persist as hydatid of Morgagni.

- _____ 23. During the development of the ***female reproductive system***, which statement is ***false***?
- a) The urogenital folds remain unfused and form the labia minora.
 - b) The cranial portion of the paramesonephric ducts form the uterine tubes.
 - c) Didelphys uterus anomalies occur when there is a complete lack of fusion of paramesonephric ducts.

d) The phallus forms the mons pubis.

e) The labioscrotal swellings form the labia majora.

_____ 24. During the development of the **male reproductive system**, which statement is **false**?

- a) Hypospadias is constantly associated with exstrophy of urinary bladder.
- b) Because of the mode of descent of testis, the ductus deferens crosses anterior to the ureter.
- c) Sertoli cells, derived from mesoderm, constitute most of the seminiferous epithelium in the fetal testis.
- d) Interstitial cells (of Leydig) begin to secrete androgenic hormones by the end of the third trimester.
- e) The epididymis, ductus deferens, ejaculatory duct, and seminal vesicle all arise from the paramesonephric ducts.

_____ 25. A structure found within the adult **female pelvis formed from the gubernaculum** is

- a) broad ligament
- b) superior ligament of the ovary
- c) medial umbilical ligament
- d) round ligament of the uterus
- e) median umbilical ligament

_____ 26. The **labia minora** arise embryologically from:

- a) phallus
- b) labioscrotal swellings
- c) sinovaginal bulbs
- d) urogenital folds
- e) paramesonephric duct

_____ 27. A patient is born with karyotype **46, XY** but has no problem with production of testosterone or müllerian inhibiting hormone. Their **vagina ends in a blind pouch** and **testes are lodged in the inguinal canal**. Which of the following is the best description of this patient:

- a) Male pseudo-hermaphrodite
- b) Female pseudo-hermaphrodite
- c) Turner's syndrome
- d) Androgen insensitivity syndrome
- e) Klinefelter syndrome

_____ 28. Which of the following inhibits the growth of the **paramesonephric duct (males)**?

- a) Testosterone
- b) Leydig Cells
- c) Dihydrotestosterone
- d) Estrogens
- e) Müllerian substances (MIFs)

_____ 29. The **epididymis, ductus deferens, ejaculatory duct, and seminal vesicle** all arise from:

- a) The Müllerian ducts
- b) The paramesonephric ducts
- c) The mesonephric ducts
- d) Leydig hormones
- e) Wolffian hormones

_____ 30. The **uterine tubes, uterus, cervix, and superior part (1/3) of the vagina** all arise from:

- a) The Wolffian ducts
- b) The paramesonephric ducts
- c) The mesonephric ducts
- d) Leydig hormones
- e) Müllerian hormones

_____ 31. What structure in the **indifferent stage** (weeks 3-6) becomes the **clitoris**?

- a) Urogenital groove
- b) Genital swelling
- c) Urethral fold
- d) Perineum
- e) Genital tubercle

_____ 32. During **development of the genital system**, which statement is **false**?

- a) The lower portion of the vagina is derived from urogenital sinus.
- b) During development of the ovaries, the primitive sex (medullary) cords degenerate.
- c) The epoophoron and paroophoron, which are important as cysts, arise from the mesonephric ducts.
- d) The gubernaculum is attached to the uterus and as the ovaries descend it becomes the gubernacular ligament of the uterus.
- e) After the sinovaginal bulbs proliferate and fuse, they form a solid core of endodermal cells called uterovaginal primordium.

